Expt No.9 Dt:

STUDENT PERFORMANCE ANALYSIS USING NUMPY AND PANDAS

Aim: To write and execute a Python program for analyzing student performance data using NumPy and Pandas.

Interactive Development Environment (IDE) Used: Jupyter NoteBook
Programming Language Used : Python

Python Concepts used:

NumPy and Pandas are important Python libraries used for numerical computing and data analysis. NumPy provides functions for efficient numerical operations. Pandas provides data structures and functions for organizing, analyzing, selecting and processing tabular data. A Pandas DataFrame is a two-dimensional table consisting of rows and columns. In addition, Python supports indexing, conditional expressions and functions for processing data.

Python Concept	Example	Description
NumPy	np.sum(), np.mean()	Calculates the sum or mean of numerical data.
Pandas	pd.DataFrame()	Creates a DataFrame from structured data.
DataFrame	df	Stores data in rows and columns.
Row-wise Operation	axis=1	Applies operation across columns for each row.
Maximum Index	idxmax()	Returns the index of the maximum value.
Location based indexing	loc[]	Access a group of rows and columns by label(s).
Conditional Selection	np.where()	Selects values based on a condition.

Python Program:

```
import numpy as np
import pandas as pd
# Marks of five students in three subjects
data = {
   "Name": ["Anu", "Balu", "Chandu", "Divya", "Eswar"],
   "Python": [85, 72, 90, 65, 78],
   "Statistics": [80, 75, 88, 70, 82],
   "Data Science": [90, 68, 92, 75, 85]
}
# Create a DataFrame
df = pd.DataFrame(data)
# Calculate total and average marks
marks = df[["Python", "Statistics", "Data Science"]]
df["Total"] = marks.sum(axis=1)
df["Average"] = marks.mean(axis=1)
# Assign result based on average
df["Result"] = np.where(
   df["Average"] >= 70, "Pass", "Needs Improvement"
)
# Display results
print("STUDENT PERFORMANCE REPORT")
print(df.round(2).to_string(index=False))
```

```
# Find the highest average
topper = df.loc[df["Average"].idxmax()]

print("\nTop Performer:", topper["Name"])
print("Highest Average:", round(topper["Average"], 2))
```

Sample Input and Output:

Input: The student marks data are predefined in the Python dictionary in the program. No keyboard input is required during execution.

Name	Python	Statistics	Data Science
Anu	85	80	90
Balu	72	75	68
Chandu	90	88	92
Divya	65	70	75
Eswar	78	82	85

Expected Output:

Result: The Python program was written and executed successfully for analyzing student performance data using NumPy and Pandas.